AHOPPA: Agentic Humanoid Operation through Perception, Planning, and Action

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https://hebbian-robotics.github.io/AHOPPA/

Abstract: Humanoid operation couples navigation, contact and completion judgments while physics continues during inference. We report results obtained in the last few weeks following frontier model releases. AHOPPA connects multimodal models to a fixed SONIC controller on a simulated Unitree G1. Cameras, a map and revisable plans guide head/wrist and locomotion commands to inspect four drawers, close open ones and return home. Across 64 episodes with four models at two reasoning efforts, GPT-6 Astra succeeded in 12/14 task attempts, taking 615 to 1492 s. Independent evaluation exposes falls, timeouts and incorrect completion claims.

Keywords: Humanoid control, multimodal agents, online evaluation

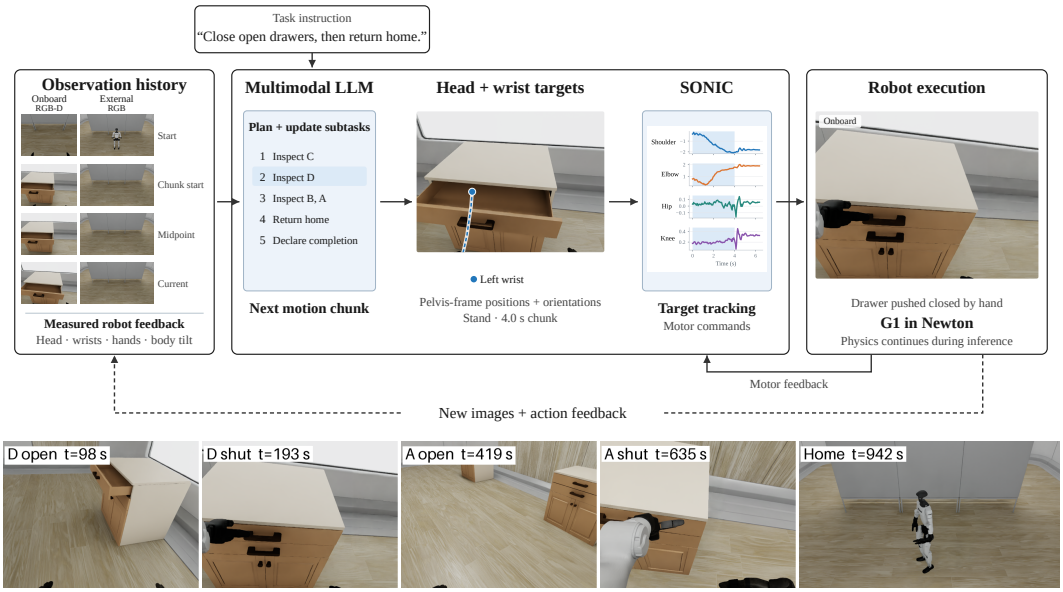


Figure 1: Top: the agent loop with recorded inputs and commands. Bottom: selected original observations from the same successful Astra (low) episode, with A and D initially open. Labels give elapsed seconds; the completion declaration followed at 955 s.

1 Online humanoid operation

Room-scale humanoid operation couples navigation, inspection, contact and stopping while physics continues during inference. We test frontier models through numerical SONIC references [1]. Prior work spans skills, code, feedback and spatial or learned action representations [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]. We measure task completion, reasoning-effort tradeoffs and failures without a task skill library.

Model	Success	Physical	Closed	Seen	False	Control	Est. $	Episode (s)	Decision (s)
Astra (low)	6/7	6	9	5	0	1/1	9.30 / 10.57	817 / 926	9.2 / 9.4
Astra (high)	6/7	6	8	6	0	0/1	7.65 / 8.78	998 / 1145	21.4 / 21.8
Sol (low)	4/7	4	4	5	1	0/1	1.72 / 1.96	829 / 940	10.4 / 10.4
Sol (high)	3/7	5	9	6	0	0/1	1.74 / 1.59	1235 / 1162	21.5 / 21.8
Opus (low)	2/7	3	5	4	2	1/1	4.78 / 6.23	762 / 1028	13.2 / 13.5
Opus (high)	3/7	3	6	3	1	1/1	7.22 / 8.36	1392 / 1563	36.1 / 34.5
Gemini (low)	0/7	0	0	1	0	0/1	1.29 / –*	737 / –*	32.3 / –*
Gemini (high)	0/7	0	2	1	0	0/1	1.59 / –*	1332 / –*	136.7 / –*

Table 1: Seven scheduled task attempts per row; all-closed controls are separate. Other counts use completed episodes. *Physical*: closed, home, upright; *Seen*: all fronts viewed; *False*: declarations without physical completion. Pairs show all / successful attempts: mean cost and episode time, median per-attempt median decision latency. Costs use published rates [19, 20, 21]. *: no successful attempt observed. API-error starts stay in denominators.

2 Interface, task and evaluation

Interface. The model receives onboard RGB-D, external RGB, robot kinematics, depth queries, a map, a revisable plan and feedback. It sends head/wrist keyframes, walking/posture commands and binary hand presets to SONIC (Fig. 1). Drawer joint positions, contacts and scores are withheld; no task skills are supplied.

Task. Four Sektion cabinets [13] stand in NVIDIA's Simple Room [14, 15]: A and B behind 2.1 m lockers [16], C and D behind home. No drawer front is visible initially. Newton [17] in Isaac Lab [18] supplies contact physics with passive drawers and flat-ground support. Eight conditions vary which drawers start 0.20 m open under the same instruction.

Scoring. Closed means within 2 cm of shut and still for 1 s. Success requires all drawers closed, a supported standing hold within 25 cm of home and the model's completion declaration. Force sensors distinguish hand pushes from other contact. We record physical completion, inspection, disturbance, falls, timeouts and API errors separately.

Protocol. After pilots, we froze the setup, prompts, evaluator, source and 1800 s budget. GPT-6.1 Sol, GPT-6 Astra, Claude Opus 5.5 and Gemini 4 Argon each attempted every condition once at low and high effort, following a seeded condition schedule.

3 Results

Task completion. Astra completed 12/14 task attempts in 615 to 1492 s, combining inspection, hand contact and return home (Fig. 1). All 43 closures were by hand; no initially closed drawer was disturbed.

Reasoning effort and efficiency. High effort lengthened decisions without consistent success gains. Astra high cost less despite more output tokens: cached reads covered 50% of its input, versus 29% at low.

Failure modes and timing. Fifteen of 51 completed task starts ended in falls, one before any model command; eight timed out and four declarations lacked physical completion. Only 17/58 completed episodes, including 7 task successes, met the 100 ms physics-lag limit; median peak lag was 192 ms. The results retain timing-limited episodes. Falls rarely followed lag spikes.

Limits. One simulated layout, drawer interaction and trial per model/effort/condition limit generalization. Friction, damping and ground support are modeling choices. Reliability, gains and transfer need repeated trials, older-model baselines and hardware experiments.

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